

DESIGN PROJECT REPORT

VR-AUGMENTED TELEOPERATED ROBOT FOR SURGERIES

Submitted by

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Under the guidance

of

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in partial fulfillment for the award of the degree of

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in

MECHATRONICS



HINDUSTAN
INSTITUTE OF TECHNOLOGY & SCIENCE
(DEEMED TO BE UNIVERSITY)

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BONAFIDE CERTIFICATE

Certified that this project report titled "**VR Augmented Teleoperated Robot for Surgeries**" is the Bonafide work of **Shaik Sameer Salahuddin (22125004), Thirisha R (22125022), Deepak M (22125060)** who did the task under my watch during the academic year 2022 – 2023.

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ABSTRACT

The project focuses on the integration of Virtual Reality (VR) into medical education, specifically in the field of surgery. Utilizing Unity software, the team has developed VR surgical simulations to enhance the learning experience for aspiring surgeons. The project addresses the limitations of traditional surgical education by providing a risk-free environment for repeated practice, immediate feedback, and skill refinement. The VR environment allows students to immerse themselves in a virtual operating room, interact with anatomical structures realistically, and practice surgical procedures without real-world consequences. The project incorporates a comprehensive approach to VR development in Unity, covering aspects such as learning the basics of Unity, choosing a VR platform, understanding C# programming, exploring Unity's XR Toolkit, creating VR projects, designing VR environments with 3D models, implementing basic VR movements, and setting up Oculus VR integration.

The Oculus Rift is selected as the hardware for VR simulations, providing a high-quality VR experience with features such as inside-out tracking, Oculus Touch controllers, and a comfortable design. The project emphasizes the importance of hardware compatibility, scene setup, and script development in Unity for successful VR implementation. VR training proves to be a valuable tool in preparing individuals for real-life surgeries, reducing surgery time, minimizing mistakes, and instilling confidence in surgical abilities. The project highlights the transformative impact of VR on surgical education, emphasizing its role in shaping highly skilled and confident surgeons. The outcome of the project demonstrates that VR-trained individuals exhibit enhanced surgical skills.

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LIST OF FIGURES

Fig.no:	Name	Pg .no
1	Software -unity	11
2	XR-Interaction	11
3	Oculus rift	13
4	Hand design	14
5	VR Environment	14
6		
7		
8		
9		
10		
11		

LIST OF TABLES

Table.no :	Title	Pg .no
1	Literature review	13
2	Specifications of Oculus Rift	14

TABLE OF CONTENT

SN.NO	TITLE	PAGE NO
1	ABSTRACT	3
2	ACKNOWLEDGMENT	4
3	LIST OF FIGURES	5
4	LIST OF TABLES	5
5		
6	INTRODUCTION	7
7	PROBLEM STATEMENT	8
8	OBJECTIVE	8
9	LITERATURE REVIEW	9
9.1	Introduction	
9.2	Overall inference	
10	METHODOLOGY	11
11	SOFTWARE	12
12	HARDWARE	13
13	DESIGN	14
14	PROJECT OUTCOME	15
15	GANTT CHART	16

INTRODUCTION

Virtual Reality (VR) has indeed revolutionized medical education, particularly in the field of surgery. Virtual reality is a simulated experience that employs pose tracking and 3D displays to create immersive, lifelike experiences for aspiring surgeons.

Traditional surgical education often involves a significant amount of time spent in lectures and observing surgeries. While these are valuable aspects of learning, they lack the depth and practical exposure necessary for developing surgical skills. VR simulations offer a revolutionary solution by enabling students to immerse themselves in a virtual operating room. This immersion allows them to interact with anatomical structures realistically and dynamically, which is essential for skill development.

One of the primary advantages of VR in surgical education is the ability to practice surgical procedures repeatedly in a risk-free environment. This controlled and forgiving space allows students to refine their skills, understand each procedure, and develop muscle memory. Unlike in the real world, where mistakes can be costly, VR allows learners to make errors, learn from them, and perfect their techniques without consequences.

VR simulations provide immediate feedback, a feature that is often lacking in traditional surgical training. In a virtual environment, students can receive real-time guidance and assessments of their performance, helping them identify areas for improvement. This accelerates the learning curve and bridges the gap between theoretical knowledge and practical application. The ability to correct mistakes on the spot enhances the quality of learning and skill acquisition.

Beyond technical skills, it gives students confidence in surgical learners. The error-tolerant nature of VR allows students to experiment, take risks, and overcome the fear of making mistakes.

PROBLEM STATEMENT

Students lacked an immersive learning experience in surgery due to the absence of teleoperated robot interactions.

OBJECTIVE

- Enhancing the learning experience with virtual reality.
- Unity software to create VR surgical simulations for skill training and error reduction.

LITERATURE REVIEW

Introduction

The rapid advancement of virtual reality (VR) and augmented reality (AR) technologies has opened up new frontiers in healthcare. These innovative technologies have the potential to revolutionize surgical education, planning, and training, offering a safe, cost-effective, and immersive way to explore anatomical regions, simulate surgical procedures, and improve patient care. This literature review provides an overview of the key findings and applications of VR and AR in the realm of oral and maxillofacial surgery.

Surgical education has significantly benefited from VR technologies, allowing for the exploration of surgical pathways and the development of surgical skills in a controlled environment. This review aims to delve into the use of VR for surgical education and the 3D reconstruction of internal organs, providing insights into viable surgical pathways and their implementation in training programs.

Overall Inference

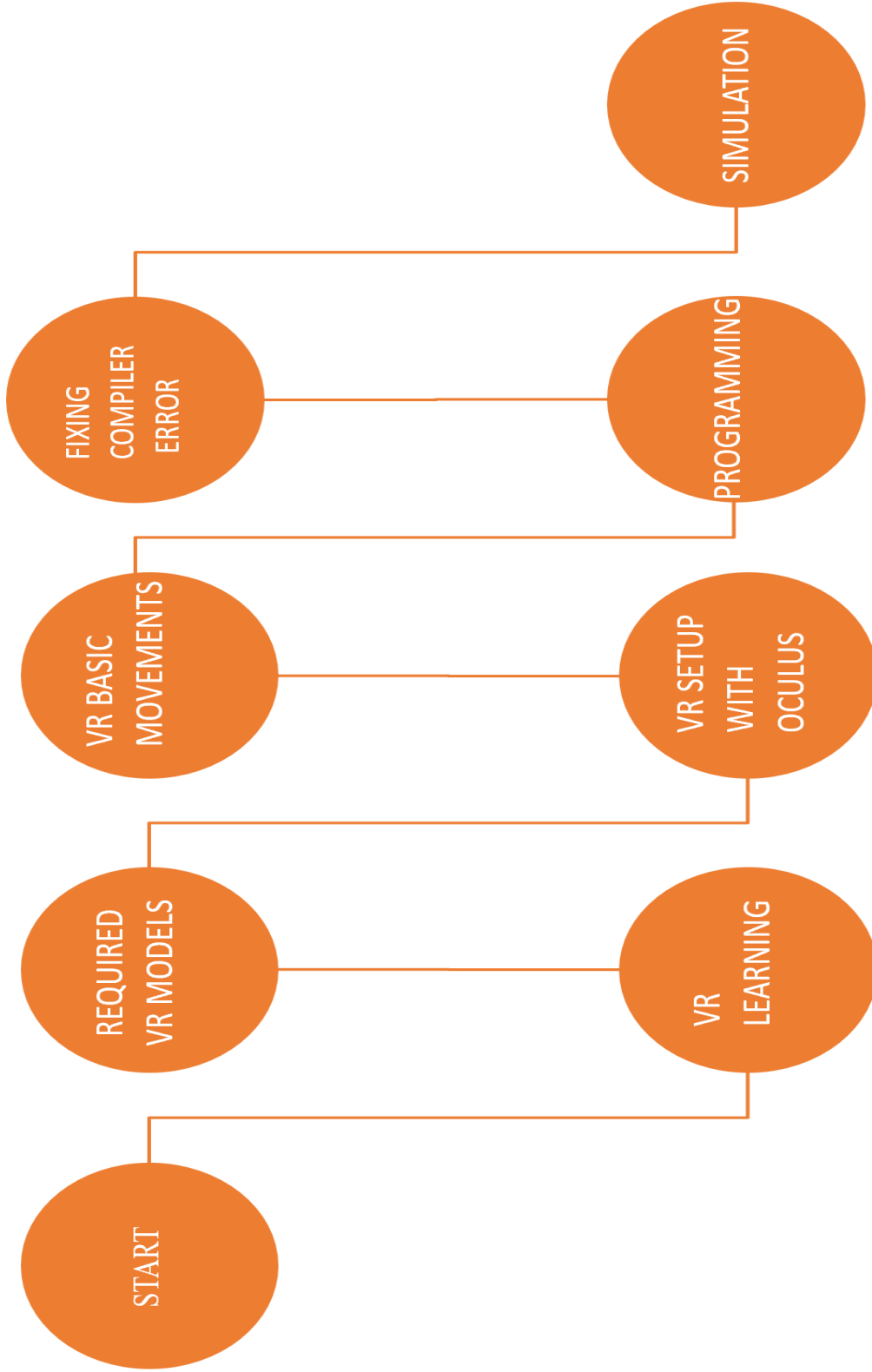
Simulation, particularly immersive virtual reality, offers an effective learning strategy to supplement real-world training, with potential applications in diverse surgical specialties. This review underlines the need for continued research and collaboration among researchers and clinicians to achieve the full potential of VR and AR in surgical education and practice. These recommendations should guide further studies and the widespread implementation of VR simulation technology in the medical field.

LITERATURE REVIEW

Table 1

S.NO	YEAR	TITLE	AUTHORS	INFERENCE
1	2022	The application of virtual reality and augmented reality in Oral & Maxillofacial Surgery	Authors: Ashraf Ayoub and Yeshwanth Pulijal	The integration of virtual and augmented reality in oral and maxillofacial surgery promises enhanced surgical planning, training, and patient outcomes.
2	2020	- Immersive VR in surgery and medical education: diving into the future-Am j Surg - Virtual reality for shoulder arthroplasty education(seminars in arthroplasty)	Schwartzberg, Steven & Cao, Caroline & De, Suvranu & Jones, Daniel	VR is reshaping medical education, particularly in shoulder arthroplasty, signaling a promising future for training and procedures.
3	2017	Virtual reality-based simulators for surgery : a systematic review	Micahel p fandler	This review confirms VR surgical simulators' potential for advancing surgical training and skills.
4	2018	Exploring the use of virtual reality in surgical education	Ntakakis, Christina Plomariti, Christos Frantzidis, Panagiotis E Antoniou and Panagiotis D Bamidis,	This study investigates the use of virtual reality in surgical education.
5	2015	Identifying Opportunities for Virtual Reality Simulation in Surgical Education: A Review of the Proceedings from the Innovation, Design, and Emerging Alliances in Surgery (IDEAS) Conference: VR Surgery. Surgical innovation.	Olasky, Jaisa & Sankaranarayanan, Ganesh & Seymour, Neal & Magee, J & Enquobahrie, Andinet & Lin, Ming & , Rajesh & Brunt, L d	This review highlights the pursuit of virtual reality simulations in surgical education at the IDEAS Conference, emphasizing their potential to enhance surgical training

METHODOLOGY



VR LEARNING

Learning VR development in Unity as a beginner can be an exciting and rewarding journey.

Here's a more detailed step-by-step start:

- **Install Unity and Set Up Environment:**

Download and install the latest version of Unity from the official website.

Create a Unity ID and familiarize yourself with the Unity Hub, which helps manage projects.

- **Understand the Basics of Unity:**

Begin with Unity's beginner tutorials to grasp fundamental concepts such as the Unity interface, scene creation, and object manipulation.

- **Choose a VR Platform:**

Decide which VR platform want to target. Some popular options include Oculus Rift, HTC Vive, and Oculus Quest for standalone VR experiences. Ensure that the PC meets the hardware requirements for the chosen VR platform.

- **Learn C# Programming:**

Unity primarily uses C# for scripting. Start learning the basics of C# programming through online tutorials, courses, or books. Understand variables, functions, and object-oriented programming principles.

- **Explore Unity's XR Toolkit:**

Unity's XR Toolkit simplifies VR development. Familiarize yourself with it to create VR interactions and set up VR settings.

Create Simple VR Projects:

Create a VR scene where can explore a basic environment. Experiment with VR interactions like teleportation, object manipulation, or simple user interfaces.

- Design for VR:

Study 3D modeling, texturing, and lighting to create an immersive VR environment.

Pay attention to the scale and proportions of objects in the VR world to ensure realism.

- Test on Real Hardware:

Regularly test VR applications on the actual hardware 're targeting. This helps identify and fix issues specific to VR platforms.

REQUIRED VR MODELS

Creating a medical VR environment in Unity is a multi-faceted process that involves designing and integrating various 3D models and elements. This immersive experience aims to provide a realistic simulation of a medical setting, complete with beds, surgical instruments, and a hospital environment. Designing the Virtual Medical Environment:

- **Choosing the Right Software:**

The journey begins with selecting a 3D modeling software like Blender, Maya, or 3ds Max. These tools allow to crafting of the 3D models needed for the medical VR environment.

- **Building the Hospital Environment:**

The hospital environment serves as the backdrop for the VR experience. Create 3D models for walls, floors, ceilings, and other architectural elements to shape the virtual world.

- **Furniture and Objects:**

Within this environment, model various objects commonly found in a medical facility, including chairs, tables, medical carts, and more.

- **Designing Medical Beds:**

Design a variety of medical beds, such as hospital beds, operating tables, and examination tables.

- **Surgical Instruments:**

Model surgical instruments that are vital to medical simulations, including scalpels, forceps, syringes, and other tools.

- **Patient Models:**

Create 3D models of patients. These models should represent human anatomy accurately and incorporate features like skin textures and animations.

- User Interface (UI):

Develop a user interface that facilitates interaction within the VR environment. This UI may include information displays, menus, and interactive buttons.

- Textures and Materials:

Apply textures and materials to 3D models to make them appear more lifelike. Utilize shaders and materials to emulate various surface properties like metal, plastic, or fabric.

- Physics:

Implement physics to ensure that objects within the VR environment behave realistically, mimicking their real-world counterparts. Unity's physics engine can be instrumental in this aspect.

- VR Integration:

Integrate 3D models into Unity and ensure compatibility with the target VR platform. Implement VR controller functionality for interactions within the environment.

- Testing and Optimization:

Rigorously test the VR medical environment to ensure that it functions as intended. Optimize 3D models and Unity scenes to guarantee optimal performance in the VR environment.

- Audio Enhancement:

Incorporate audio elements into the VR environment, such as heartbeats, machine sounds, and voiceovers. These elements add depth and immersion to the overall experience.

Creating a medical VR environment in Unity is a complex and time-intensive endeavor. It often necessitates collaboration with 3D artists, developers, and subject matter experts in the medical field to ensure accuracy and realism. Nevertheless, with careful planning and attention to detail, the result can provide an invaluable training and simulation tool for medical professionals.

VR BASIC MOVEMENTS

In Unity software, implementing VR basic movements involves using the Unity game engine along with specific VR development tools or plugins, such as those provided by Oculus, SteamVR, or other VR platforms. Here's a general overview of how one might implement some basic movements in a Unity VR project:

- **Set Up VR Environment:**

Install the necessary VR development package for the target platform (e.g., Oculus Integration for Oculus Rift, SteamVR for HTC Vive).

- **Create a VR Camera Rig:**

Set up a VR camera rig that represents the player's viewpoint in the virtual world. Unity's XR system includes prefabs for camera rigs specific to different VR platforms.

- **Teleportation:**

Use the teleportation system provided by the VR development package. For example, Oculus provides the OVRInput API, and SteamVR provides the Teleport script. These allow to detection of controller input and move the player to a desired location.

- **Free Roaming / Natural Locomotion:**

Implement the Unity XR Input system to capture the user's physical movements. can use the Input. Get the Axis method to read the user's movement on the x, y, and z axes and apply those movements to the VR camera rig.

- **Trackpad/Joystick Locomotion:**

Use the Unity Input System to capture joystick or trackpad input. Translate the input values into movement for the VR camera rig. might want to incorporate physics to simulate acceleration and deceleration for a smoother experience.

- Snap Turning / Smooth Turning:

Implement turning using the XR input system or the input system provided by the VR development package. For snap turning, set predefined rotation increments. For smooth turning, interpolate the rotation based on the input.

- Climbing and Grabbing:

Use the VR development package's input system to detect grabbing or climbing gestures.

Attach colliders and scripts to interactive objects to handle the interaction. The XR Interaction Toolkit in Unity is a useful resource for implementing these features.

- Testing and Optimization:

Regularly test VR movements to ensure they are comfortable and intuitive. Optimize performance by considering factors like frame rate and minimizing motion sickness.

VR SETUP IN OCULUS

Setting up Oculus VR in Unity with controllers involves several steps to enable interaction and create immersive VR experiences. Set up Oculus VR with controllers in Unity:

- Hardware and Software Requirements:

Ensure have a compatible Oculus VR headset (e.g., Oculus Rift, Oculus Quest, or Oculus Quest

Verify that the computer meets the necessary hardware specifications for Oculus VR.

Download and install the Oculus software on the computer.

- Install Unity:

Download and install Unity, a game development engine compatible with Oculus VR. Use a

Unity version that supports Oculus integration (Unity 2020 or later).

- Oculus Integration Package:

Import the Oculus Integration package into the Unity project. This package provides essential tools and prefabs for Oculus VR development, including controller support.

- Create a New Unity Project:

Start a new Unity project, selecting the 3D template. Configure project settings to match Oculus VR specifications.

- Oculus Camera Rig:

In the Unity scene, add the **OVRCameraRig** prefab provided by the Oculus Integration package. This prefab sets up the VR camera and controllers.

- VR Interactions:

Add objects and assets to the scene that users can interact with in VR. Using Unity's physics system for this. Implement VR interactions for grabbing, manipulating, and interacting with objects using Oculus controllers. Using the **OVRGrabber** script for this purpose.

- User Interface (UI):

Design and integrate user interfaces optimized for Oculus VR. These interfaces should work well with VR headsets and controllers. Implement gaze-based or controller-based UI interactions for menu systems and user feedback.

- Locomotion and Movement:

Choose a locomotion method, such as teleportation or smooth locomotion, and implement it in a way that minimizes motion sickness. Set up teleportation points or create a walking system for user movement, considering Oculus comfort guidelines.

- Controller Inputs:

Utilize the Oculus Input API to capture controller input, such as button presses, thumbstick movements, and trigger interactions. Implement controller haptic feedback for tactile responses in the VR environment.

- Performance Optimization:

Optimize VR experience to maintain a smooth frame rate for Oculus VR. Apply techniques like occlusion culling, level of detail (LOD), and asset optimization.

- Testing and Debugging:

Regularly test the Oculus VR application on an Oculus headset to ensure it functions correctly.

Debug any issues specific to Oculus VR, including controller interactions and camera setup.

- Audio for Immersion:

Use 3D audio to create an immersive soundscape in the Oculus VR experience. Position audio sources accurately in the 3D space to match the visual elements.

- Safety and Comfort:

Prioritize user safety and comfort in the Oculus VR experience. Avoid actions that can cause motion sickness, such as intense movements. Offer options for user comfort, like adjustable settings for movement and graphics quality.

PROGRAM FOR VR

Creating C# scripts in Unity is a fundamental aspect of VR development.

- Open Unity:

Launch the Unity Editor on the computer.

- Create or Open a Project:

Either start a new Unity project or open an existing one, and add C# scripts.

- Access the Assets Panel:

In the Unity interface, locate the "Assets" panel. It's typically in the lower part of the window, within the "Project" tab.

- Create a New C# Script:

To create a new C# script, right-click in the "Assets" panel. A context menu will appear.

- Choose "Create" > "C# Script":

In the context menu, hover over "Create" to reveal a submenu. Select "C# Script" from the submenu.

- Name the Script:

After selecting "C# Script," a new script file will appear in the "Assets" panel, with a default name like "NewBehaviourScript." Rename it to something relevant to the script's purpose.

Right-click on the script and select "Rename" to change its name.

- Open the Script in the Preferred IDE:

To start working on the script, double-click on the script file in the "Assets" panel. This will open the script in the preferred integrated development environment (IDE).

- Editing the C# Script:

In the chosen IDE, can write, modify, and debug C# code for the Unity project. This is where to implement the functionality and behavior of VR objects, characters, interactions, etc...

- Save the Script:

Write or modify the script, and save the changes within IDE. Unity will automatically recognize and recompile the script when saving it.

- Attach the Script to VR Objects:

To make the script interact with VR objects in the Unity scene, typically attach it as a component to the relevant objects. Do this by dragging and dropping the script file onto the object in the Hierarchy panel.

- Testing and Debugging:

Use Unity's built-in debugging tools to test and debug scripts. Set breakpoints, inspect variables, and monitor the behavior of scripts in real time.

Creating and working with C# scripts in Unity is a fundamental skill for game development.

By following these steps, create and manage scripts to bring the Unity projects to life and add interactivity and functionality to VR objects.

FIXING COMPILER ERRORS

When working with VR development in Unity, some common compiler errors or issues may arise, but it's important to note that the Unity development environment itself doesn't have a compiler in the traditional sense. Unity uses the Mono runtime, which includes a just-in-time (JIT) compiler to convert C# code into machine code at runtime.

That being said, when developing VR applications in Unity, some common issues that may lead to errors or unexpected behavior include:

- **Compatibility Issues:**

VR platforms often have specific requirements and SDK versions. Ensure that the Unity version and VR platform SDK are compatible.

- **Hardware Requirements:**

VR development may require specific hardware capabilities. Ensure that the development machine meets the VR platform's hardware requirements.

- **Incorrect SDK Integration:**

If targeting a specific VR platform (e.g., Oculus Rift, HTC Vive), ensure that have integrated the corresponding SDK correctly. Issues with SDK integration can lead to runtime errors.

- **Scene Setup:**

VR scenes often require specific configurations for cameras, player controllers, and interactions. Incorrect scene setup can lead to unexpected behavior.

- **Script Errors:**

Check C# scripts for syntax errors, typos, or logical issues. Unity's console window will provide information about any script-related errors.

- **Asset Import Issues:**

Improperly imported assets, such as 3D models or textures, can cause runtime issues. Check the import settings and make sure they are compatible with the VR platform.

- **Plugin Compatibility:**

If using third-party plugins or assets, ensure that they are compatible with the Unity version and the VR platform.

- **Memory and Performance:**

VR applications demand optimized performance. Pay attention to memory usage, frame rates, and GPU performance. Use Unity's profiling tools to identify performance bottlenecks.

- **Unity Version:**

Sometimes, using an outdated or incompatible Unity version may lead to issues. Ensure are using a stable version recommended for VR development.

To troubleshoot these issues, Unity's console window and error messages are valuable resources.

When working with Oculus headsets in Unity, there are specific considerations and potential issues that developers may encounter. Here are some common factors and challenges related to Oculus headset integration with Unity:

- **Oculus Integration Package:**

Import the Oculus Integration package. Ensure that one has the correct version of the Oculus Integration package compatible with the Unity version.

- **SDK Compatibility:**

Oculus regularly updates its SDK, and Unity might release new versions. Ensure that the Oculus Integration package and Unity version are compatible with the latest Oculus SDK to take advantage of new features and improvements.

- **Tracked Objects and Controllers:**

Oculus headsets come with tracked controllers. Make sure to properly set up and map these controllers in the Unity scene. Unity's Input System can help handle input from Oculus controllers.

- **Performance Optimization:**

VR applications demand high performance. Optimize Unity project for smooth frame rates and low latency to avoid motion sickness. Utilize Oculus Performance HUD and Unity Profiler to identify performance bottlenecks.

- **Stereo Rendering and Cameras:**

Oculus headsets rely on stereo rendering for a 3D effect. Ensure that Unity cameras are configured correctly for stereo rendering, and adjust the IPD (Interpupillary Distance) to enhance the user experience.

- **Oculus Avatar SDK:**

If want to include avatars in VR applications, consider using the Oculus Avatar SDK. This SDK allows to integration of realistic avatars with hand tracking, which can enhance the sense of presence in VR.

- **Platform-Specific Deployment:**

When deploying a VR application, be aware of platform-specific requirements for Oculus Store submission if plan to distribute the app through the Oculus Store.

- **Firmware and Drivers:**

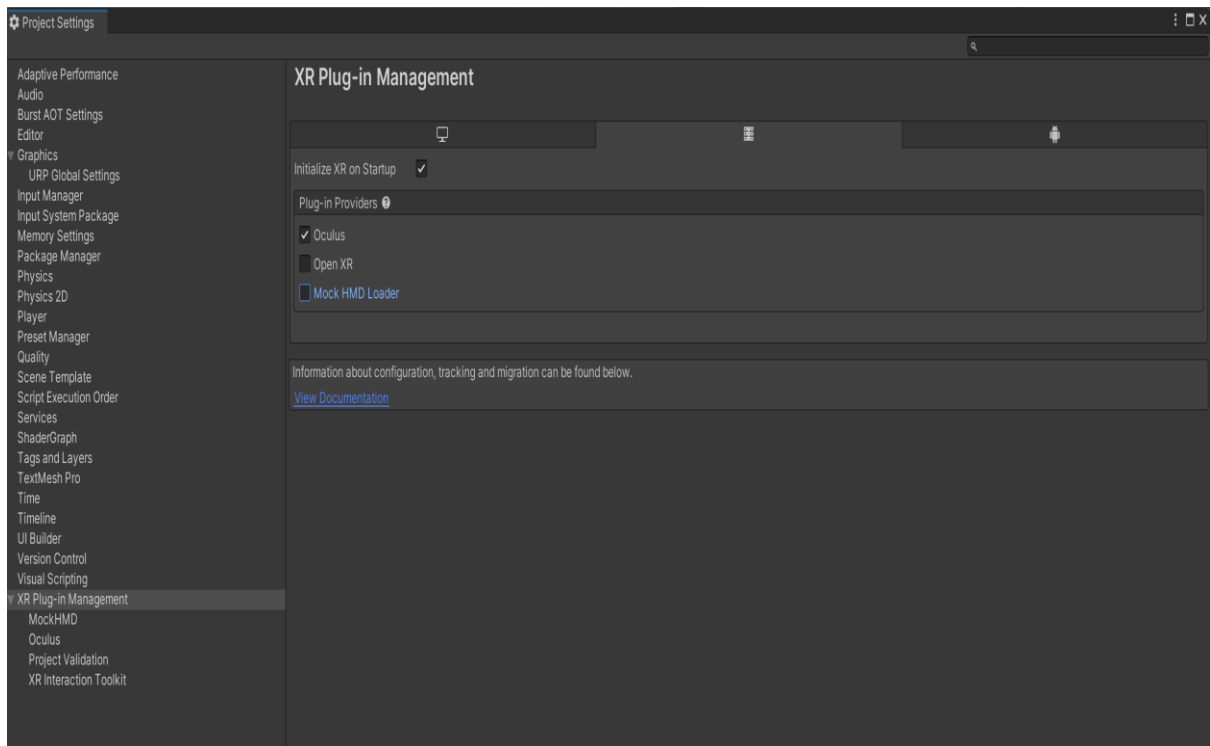
Keep Oculus firmware and drivers up-to-date to ensure compatibility with the latest features and improvements. Oculus periodically releases updates to enhance the user experience. Regularly check the Oculus Developer Documentation and Unity forums for updates, best practices, and solutions to common issues.

SOFTWARE



The project utilizes Unity with URP version 2023.v3

Fig 1



Used Oculus plugin for interaction simulation

Fig 2

WHAT IS UNITY?

Unity is a powerful and widely used game development engine and platform that enables developers to create 2D, 3D, augmented reality (AR), and virtual reality (VR) experiences. Unity is known for its flexibility and ease of use, making it accessible to both beginners and experienced developers. In the context of VR (Virtual Reality), Unity provides a comprehensive set of tools and features to create immersive and interactive virtual environments. Developers can use Unity to build VR applications for various platforms, including popular VR headsets like Oculus Rift, HTC Vive, PlayStation VR, and others.

Unity stands out as a powerful game development engine and platform, particularly in the realm of Virtual Reality (VR). Its strength lies in fostering cross-platform development, enabling VR applications to run seamlessly on various platforms, thus broadening their accessibility. The extensive Unity Asset Store serves as a treasure trove for developers, offering a plethora of assets, tools, and plugins to enhance and streamline VR projects. With robust physics and rendering systems, Unity contributes to crafting realistic and immersive VR experiences. The platform's scripting capabilities, primarily utilizing C#, cater to a diverse range of developers, allowing for the creation of intricate and interactive VR applications.

Unity's support for VR-specific Application Programming Interfaces (APIs) facilitates the seamless integration of hardware features like motion tracking and haptic feedback, enhancing the overall user experience. The Unity XR Framework adds another layer of flexibility, empowering developers to build applications that can seamlessly transition between various VR and AR devices with minimal code changes.

Unity with URP version 2023.v3

Unity continues to advance its visionOS beta program, enhancing integration with PolySpatial technology for a Shared Space on Apple Vision Pro. In Unity 2023, there's extended support for Foveated Rendering API, initially for PlayStation VR2 and now for Meta Quest 2 and other devices. This integration with Oculus XR and OpenXR plug-ins facilitates more immersive XR experiences. The new Split Graphics Jobs threading mode, initially for DX12 platforms and PlayStation 4, extends to PlayStation 5 in Unity 2023.v3, enhancing CPU performance for rendering thread-bound applications. Adaptive Vsync presentation mode support for PS5 in 2023.2 reduces Vsync-related frame stutter, with plans to extend it to Xbox Series and Vulkan desktop platforms in 2023.v3. Unity 2022 LTS laid the foundation for cross-play multiplayer, exemplified by the Megacity Multiplayer sample. In 2023.v3, Unity aims to simplify creation workflows, focusing on Editor improvements, dedicated server management, code stripping, deployment tools, Play mode testing, and educational content. Render Graph integration into Universal Render Pipeline (URP) optimizes GPU memory and bandwidth utilization, enhancing performance and energy efficiency. URP render passes now use the Render Graph framework, emphasizing scalability and performance on diverse platforms. Enhancements to artist workflows include Mipmap streaming debug views for SRPs, Shader Graph improvements, and enhanced URP support for VFX Graph. Unity prioritizes stability and usability in light baking, providing a seamless lighting experience. Unity continues to consolidate the foundations of the Data-Oriented Technology Stack (DOTS) Entity Component System (ECS), introduced in Unity 2022 LTS. Quality-of-life improvements enhance workflow efficiency, offering faster entity creation and improved documentation.

Overall, Unity 2023.v3 focuses on performance, quality, and enhanced workflows, offering developers tools to create more immersive, high-fidelity experiences across a variety of platforms.

HARDWARE - OCULUS RIFT

Fig 3



The Oculus Rift offers a compelling set of specifications for an immersive virtual reality (VR) experience. Featuring a display resolution of 2160 x 1200 pixels (1080 x 1200 per eye) with a 90 Hz refresh rate, the headset provides clear and responsive visuals for users. The Oculus Insight technology facilitates inside-out tracking, eliminating the need for external sensors and contributing to a more straightforward setup process. The inclusion of Oculus Touch controllers, known for their improved ergonomics and finger tracking, enhances the user's ability to interact naturally with the virtual environment. The headset's field of view is approximately 110 degrees, contributing to a more immersive experience. The single-cable system for video and USB connectivity, along with the compatibility with VR-ready gaming PCs, ensures that users can easily integrate the Oculus Rift S into their existing setups. Comfort is prioritized with the Halo headband design, offering improved weight distribution and adjustability.

The integrated audio system with built-in speakers adds to the overall convenience, providing an audio solution without the need for additional peripherals. The Passthrough+ feature allows users to glimpse the real world without removing the headset, adding a practical dimension to the user experience. It is also compatible with Oculus Link, enabling users to connect to a PC and access a broader range of VR content. As VR technology is dynamic and continually evolving, it is advisable to check the official Oculus website or other reliable sources for the latest information on Oculus Rift devices and any potential updates to their specifications.

SPECIFICATIONS OF OCULUS RIFT

Table 2

Audio	Integrated 3D audio headphones
Display	PenTile OLED
Ports	HDMI 1.3, USB 3.0, USB 2.0
Refresh Rate	90 Hz
Resolution	2160x1200 (1080x1200 per eye)

DESIGN

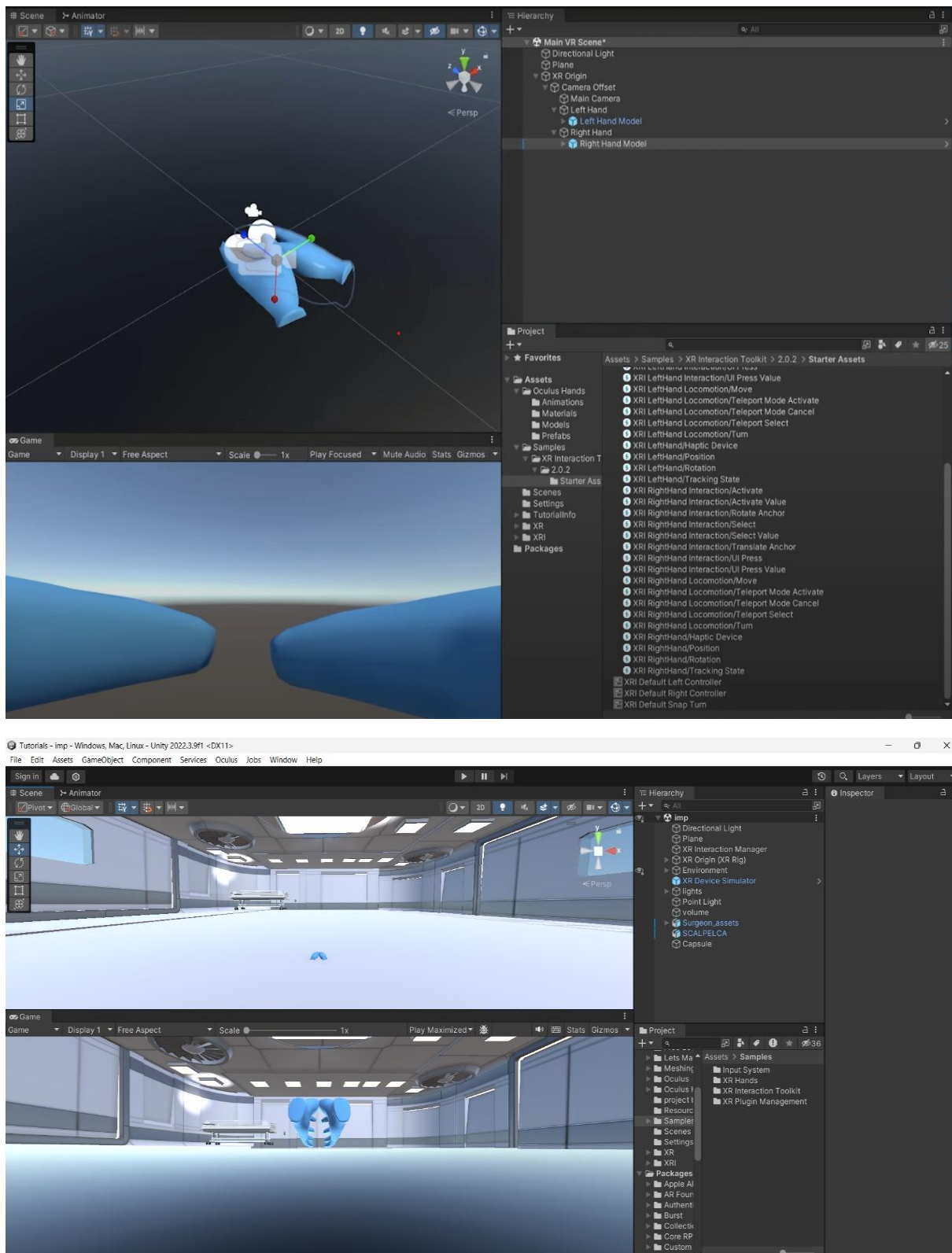


Fig 4,5

The team has designed functional hands with grab, interaction controls, and a VR Environment.

ASSET USED

Unity ships with multiple standard assets. These are collections of assets that are widely used by most Unity customers. These are 2D, Cameras, Characters, Cross Platform Input, Effects, Environment, Particle Systems, Prototyping, Utility, Vehicles.

Unity transfers standard assets into and out of projects using Unity packages.

Importing Standard Assets: See Asset Packages for information about installing (importing) standard assets, as well as sharing custom packages between projects and via the asset store.

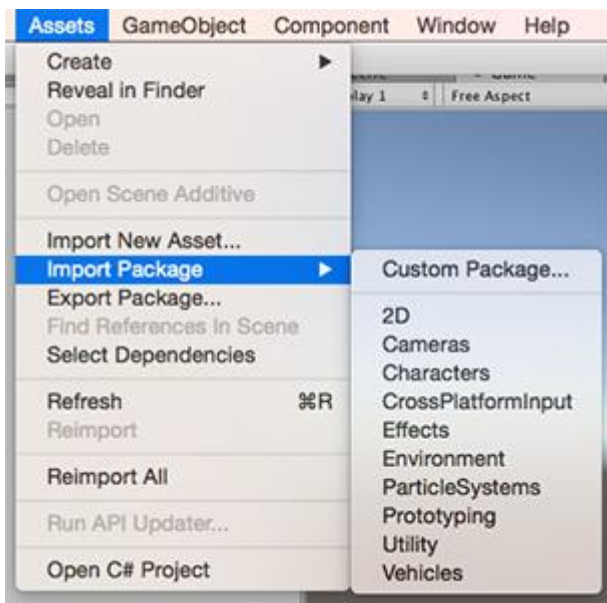


Fig 6

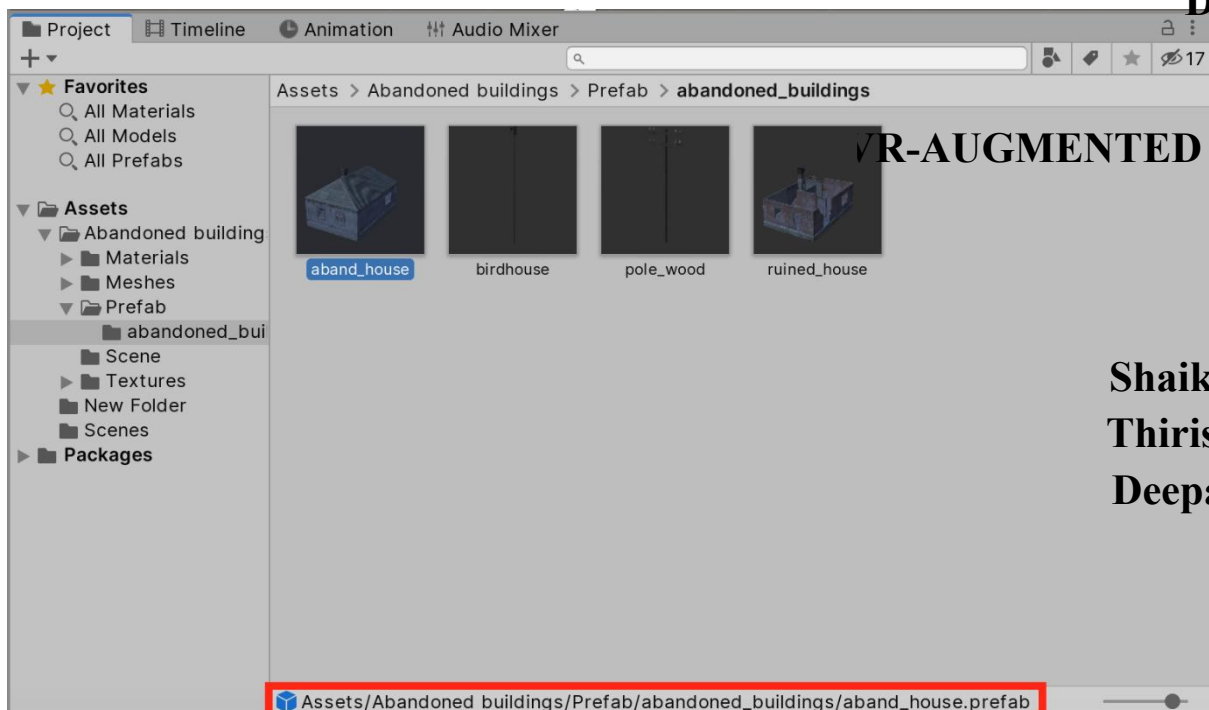
PREFABS

Prefabs are a special type of component that allows fully configured GameObjects to be saved in the Project for reuse. These assets can then be shared between scenes, or even other projects without having to be configured again. This is quite useful for objects that will be used many times, such as platforms. A major benefit to Prefabs is that they are essentially linked copies of the assets that exist in the Project window. This means that changes made or applied to the original Prefab will be propagated to all other instances. This makes fixing object errors, swapping out art or making other stylistic changes very efficient.

Prefabs are created automatically when an object is dragged from the Hierarchy into the Project window.

Prefabs look quite similar to other objects that appear in the Project window. However, when selected, their file type will be a *.prefab. When the Prefab is selected, the Inspector displays all of the components that were configured on the original object.

Fig 8



DESIGN PR

VR-AUGMENTED TELEOP

Sub

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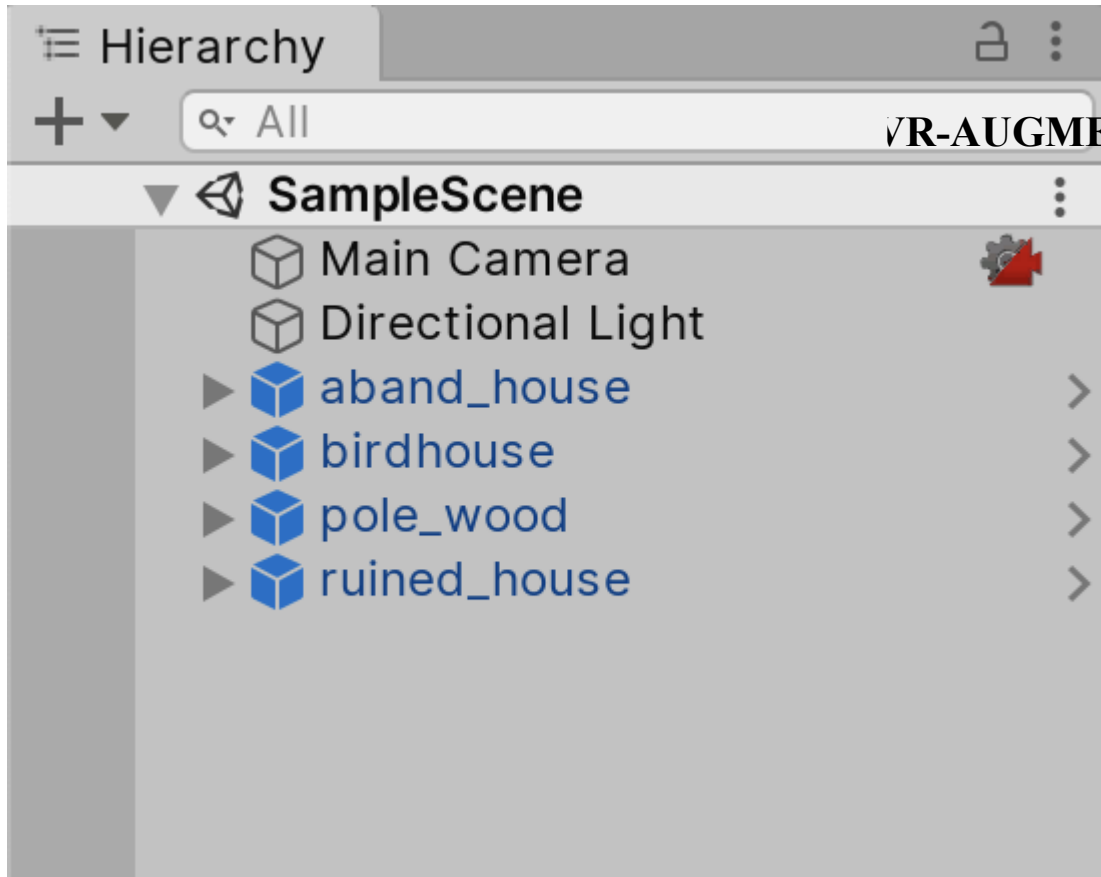
Under

Guide :

Co-guide:

When prefabs are present in the Hierarchy, they're represented with blue text and a blue cube.

Fig 9



DESIGN P

VR-AUGMENTED TELEOP

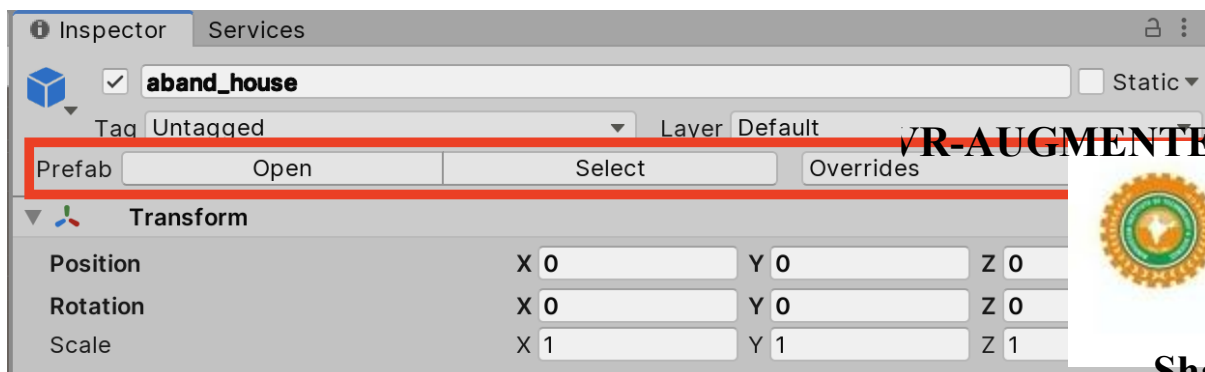
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Guide :
Co-guide:

When you convert an object to a Prefab, a new set of buttons is added above the Transform values.

Fig 9



DESIGN

VR-AUGMENTED TELEOP



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Like any other asset in Unity, Prefabs can be edited after they are created. A Prefab can be edited on a per-object basis, where a single instance of a Prefab is changed for an individual purpose, or these changes can be applied to all instances of the Prefab.

- The Select button highlights the original Prefab in the Projects folder. This is helpful for projects that have become large with many different Prefabs spread throughout many folders.
- The Revert button removes all changes applied to a Prefab that do not match with the Prefab in the Project folder.
- The Overrides dropdown will show all the changes that have been made to the prefab. At the bottom, you will see two buttons **Revert All** and **Apply All**. You can also click on each change to review, revert, and apply.
- Revert All: Will revert all the changes in the overrides list.
- Apply All: The button takes all the changes made to the selected Prefab and applies them to all other existing instances, as well as to the original in the Project window.

PROJECT OUTCOME

VR-trained individuals show enhanced surgical skills compared to non-VR trainees, with faster procedural completion, higher task scores, and improved device accuracy, achieving significant proficiency in a shorter time frame, preparing them for surgery.

People who used VR to practice surgery got good at it. They learned how to do surgeries more accurately, which is super important for the health of the patients.

Because of their VR training, these people can do surgeries more quickly. This is great because it means patients spend less time under anesthesia, reducing the risk of complications.

VR training helped people become more precise in handling surgical tools. That means they make fewer mistakes during surgeries, making the procedures safer.

The goal of the project was to prepare people for real-life surgeries. Thanks to VR training, these individuals are well-prepared and confident when they step into the operating room.

this project has shown that VR training is a way to make people better at surgery. It helps them do surgeries more quickly and accurately, reduces mistakes, and prepares them for real-life surgeries. Plus, they learn extra tech skills that can help in many areas, making the future of healthcare safer, faster, and more high-tech. Our team has acquired technical skills, including software like Unity and programming languages. VR is reshaping surgical education by offering immersive and enriched learning experiences that lead to highly skilled and confident surgeons. It equips medical students with the technical expertise, problem-solving abilities, and resilience required for success in the field of surgery. This transformation in medical education not only benefits individual learners but also has the potential to elevate the overall standard of healthcare by producing competent and proficient medical professionals.

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